# Practical Task №1 — Calculation of Bias Networks for a Bipolar Transistor Stage

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**Variant (list number $N$):** $6$

## 1. Initial Data

| Parameter | Value |
|---|---|
| $n_1$ | $6$ |
| $n_2$ | $12$ |
| Transistor | **2N5401** (PNP) |
| Supply voltage | $V_{CC} = n_2 = 12\ \text{V}$ |

## 2. Transistor Parameters

Datasheet values (Semtech 2N5401, $T_a = 25\ ^\circ\text{C}$) used in this work:

| Parameter | Symbol | Value |
|---|---|---|
| Collector–emitter voltage (max) | $-V_{CEO}$ | $150\ \text{V}$ |
| Collector current (max) | $-I_C$ | $600\ \text{mA}$ |
| Power dissipation (max) | $P_{tot}$ | $625\ \text{mW}$ |
| DC current gain ($-V_{CE} = 5\ \text{V}$, $-I_C = 10\ \text{mA}$) | $h_{FE}$ | $60 \dots 240$ |
| Collector–emitter saturation voltage ($-I_C = 10\ \text{mA}$, $-I_B = 1\ \text{mA}$) | $-V_{CE(sat)}$ | $\le 0.2\ \text{V}$ |

Pinout (TO-92): 1 — Emitter, 2 — Base, 3 — Collector.

SPICE model: **2N5401C** (Cordell Audio), simulated in KiCad 9 (ngspice).

## 3. I–V Characteristics

The characteristics were simulated within the safe operating area:

$$|V_{CE}| \le V_{CC} = 12\ \text{V} \ll 150\ \text{V}, \qquad |I_C| \le 21\ \text{mA}, \qquad P \le 12\ \text{V} \cdot 21\ \text{mA} \approx 0.25\ \text{W} < 625\ \text{mW}$$

### 3.1 Input Characteristic

DC sweep of $V_{EB}$ from $0$ to $0.8\ \text{V}$ with a $5\ \text{mV}$ step, at $-V_{CE} = 0\ \text{V}$ and $-V_{CE} = 6\ \text{V}$.

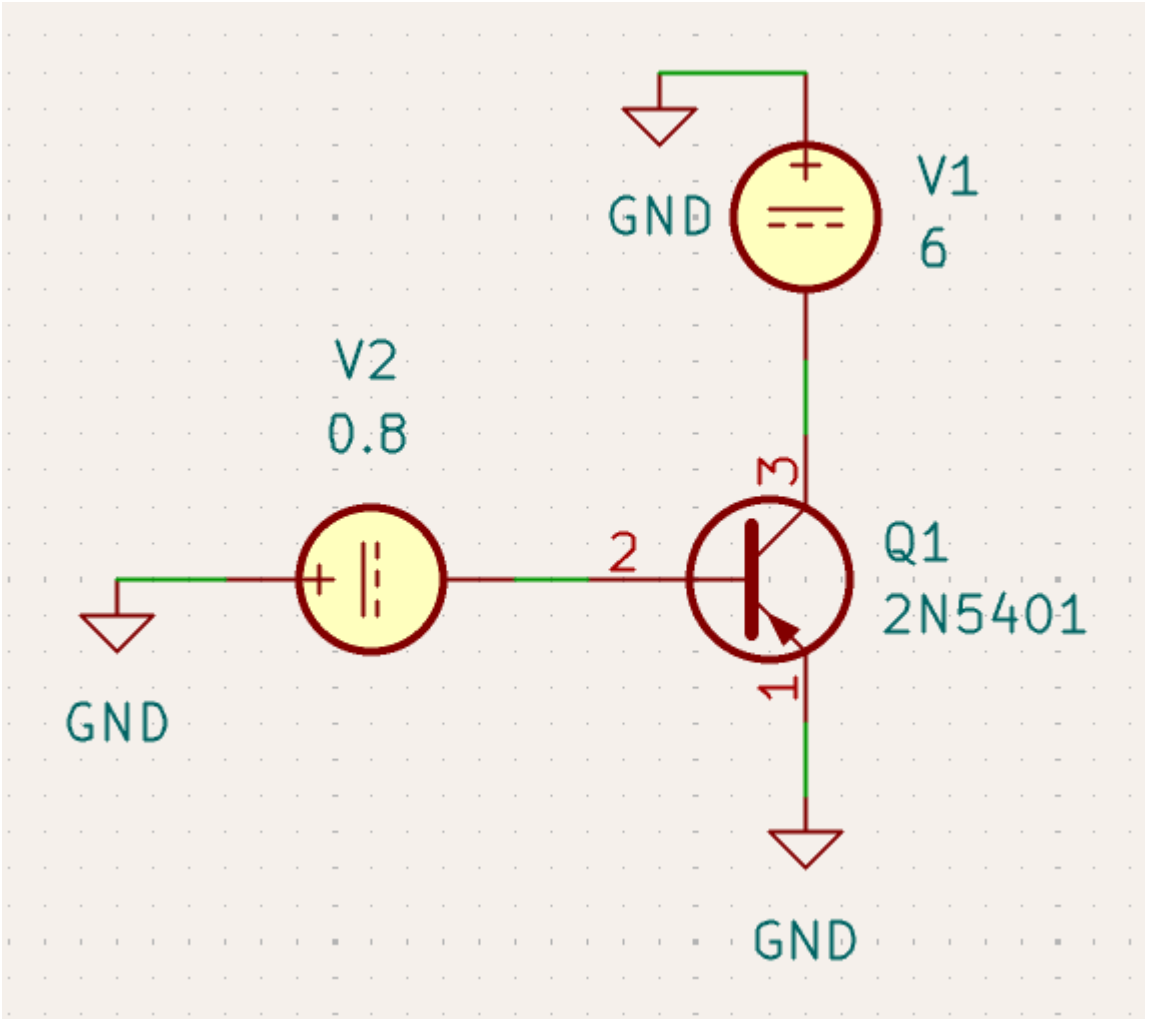



*Fig. 1. Circuit for simulating the input characteristic*

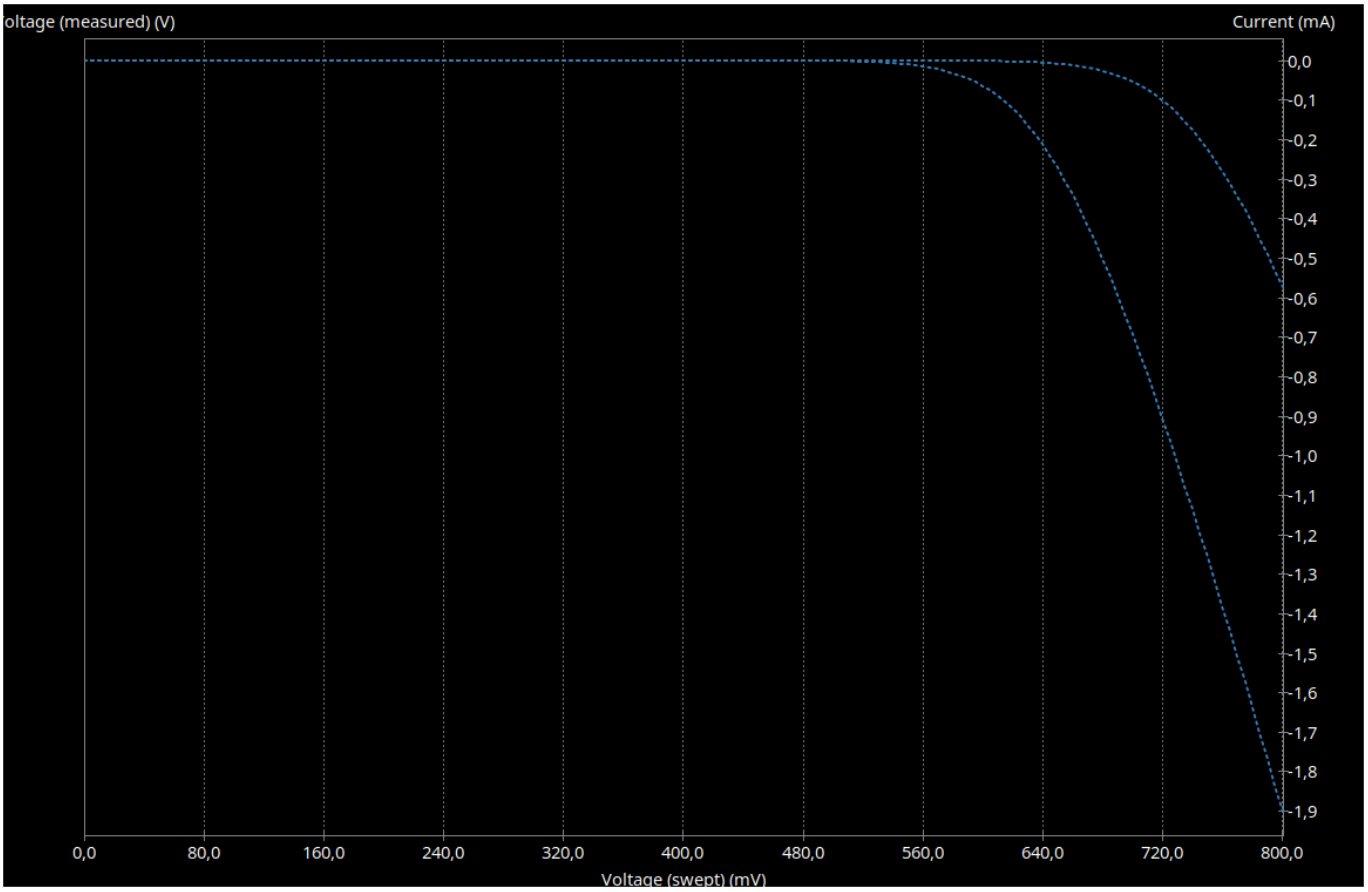


*Fig. 2. Input characteristic $I_B = f(V_{EB})$. The left curve corresponds to $-V_{CE} = 0\text{V}$, the right one to $-V_{CE} = 6\text{V}$.*

The base current is negligible below about $0.55\ \text{V}$ and rises steeply between $0.6$ and $0.75\ \text{V}$.

## 3.2 Output Characteristics

DC sweep of $-V_{CE}$ from $0$ to $12\ \text{V}$ (step $5\ \text{mV}$), with base current $I_B = 0 \ldots 100\ \mu\text{A}$ (step $20\ \mu\text{A}$).

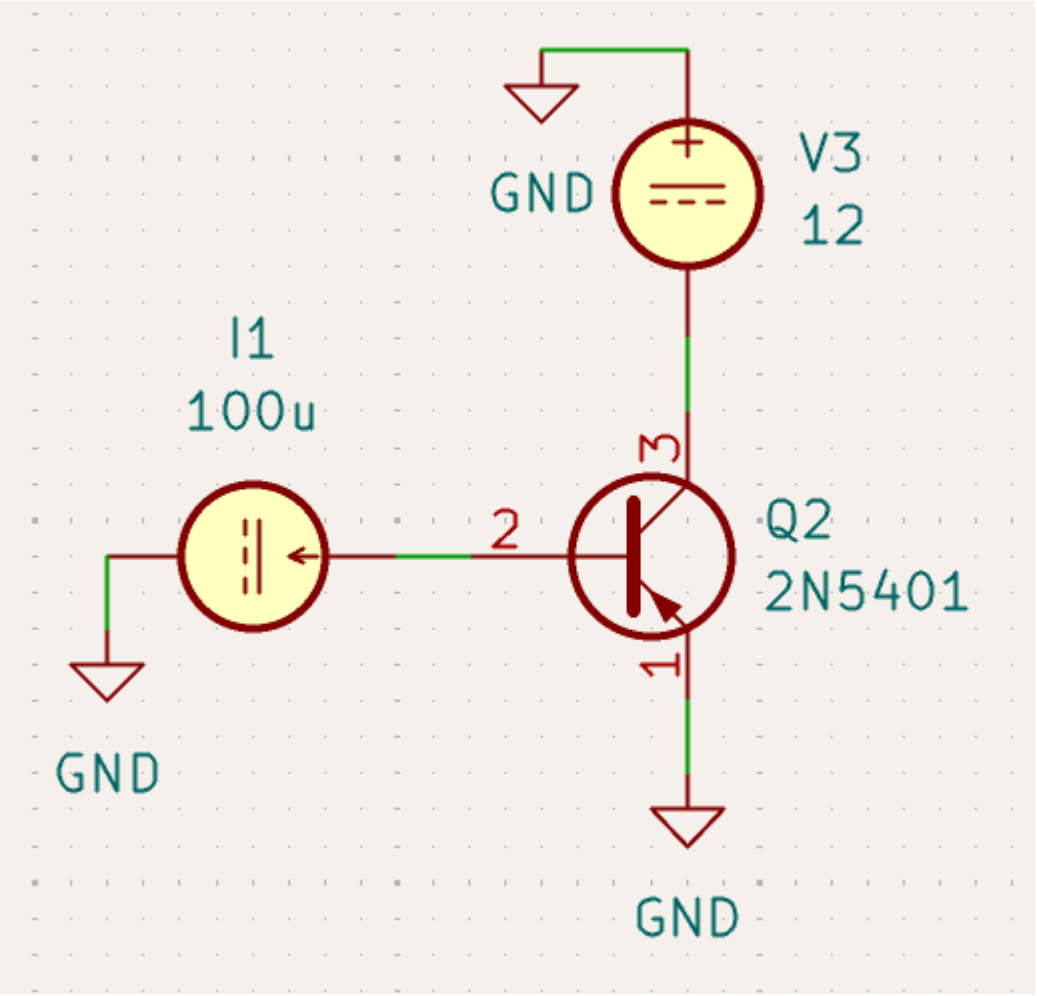



*Fig. 3. Circuit for simulating the output characteristics*

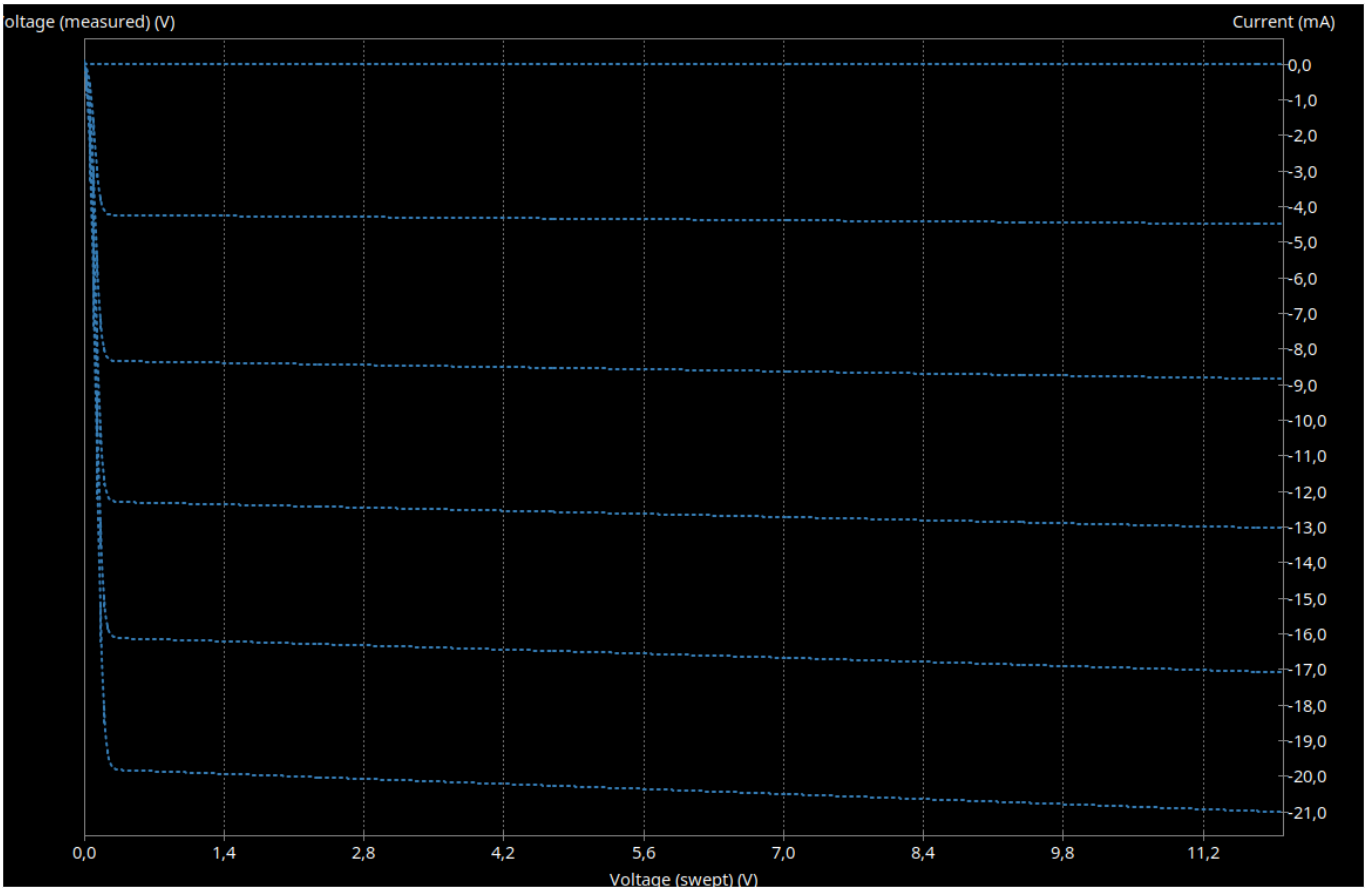



*Fig. 4. Output characteristics $I_C = f(V_{CE})$*

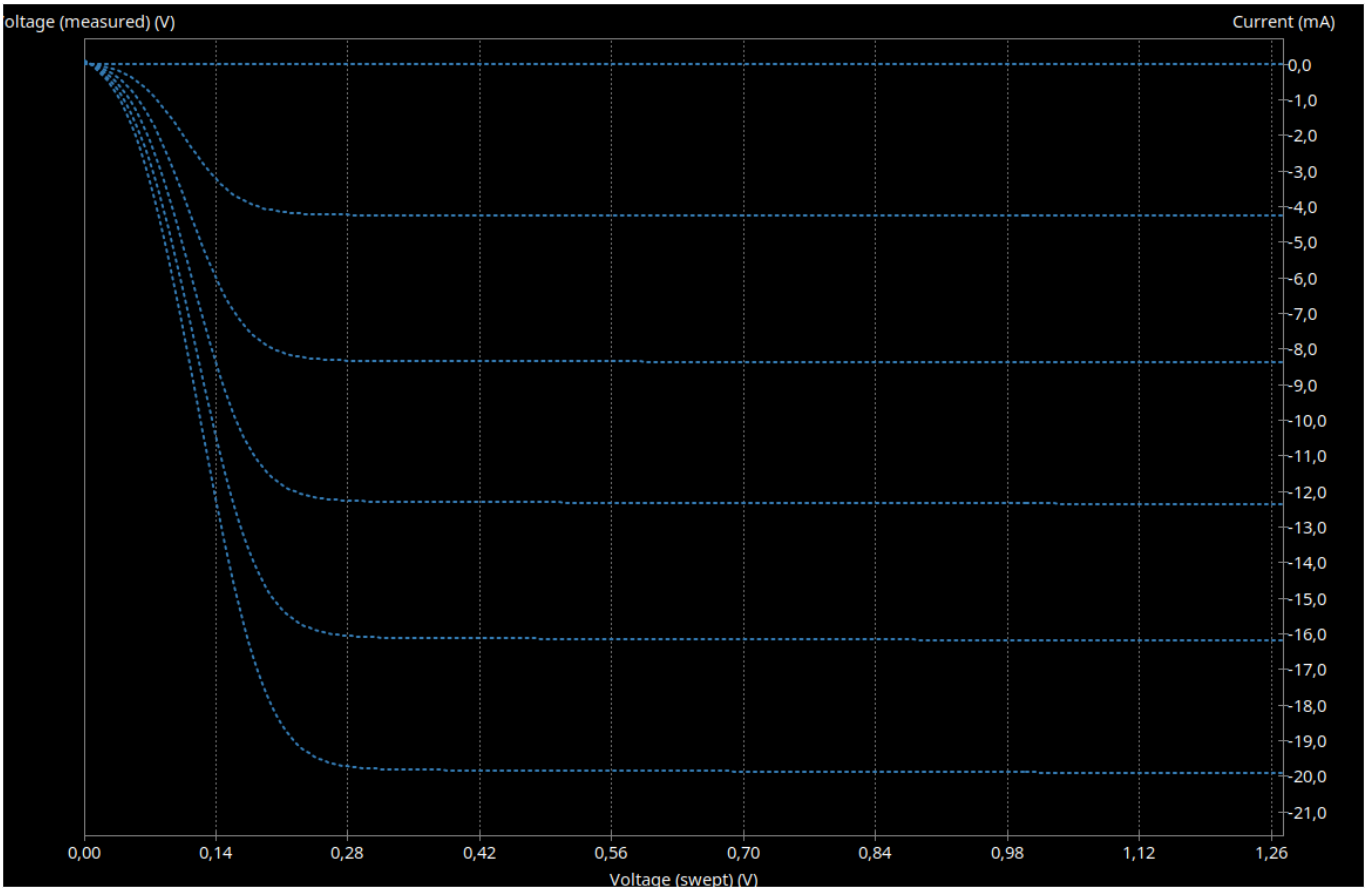



*Fig. 5. Output characteristics, saturation region (zoomed)*

| $I_B$, $\mu$A | 20 | 40 | 60 | 80 | 100 |
|---|---|---|---|---|---|
| $\lvert I_C \rvert$ at $-V_{CE} = 6$ V, mA | $\approx 4.3$ | $\approx 8.5$ | $\approx 12.5$ | $\approx 16.5$ | $\approx 20.5$ |

The saturation region ends at about $\lvert V_{CE} \rvert \approx 0.25 \ldots 0.3$ V. In the active region the curves are almost flat. The current gain is $\beta \approx 205 \ldots 215$, which lies within the datasheet range $h_{FE} = 60 \ldots 240$.

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## 4. Operating Point

| Parameter | Value | Justification |
|---|---|---|
| $V_{CE}$ | $V_{CC}/2 = 12/2 = 6$ V | Middle of the load line: maximum symmetric output swing |
| $I_C$ | $8.5$ mA | On the simulated $I_B = 40$ $\mu$A curve |
| $I_B$ | $40$ $\mu$A | From the output characteristics |
| $\beta$ | $I_C / I_B = 8.5$ mA$/40$ $\mu$A $\approx 212$ | From the output characteristics |
| $V_{EB}$ | $0.69$ V | From the input characteristic |

Check against the limits:

$$P = V_{CE} \cdot I_C = 6 \cdot 8.5 \text{ mA} = 51 \text{ mW} \ll 625 \text{ mW}, \qquad V_{CE} = 6 \text{ V} \gg V_{CE(sat)} = 0.2 \text{ V}$$

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## 5. Calculation of Bias Resistors

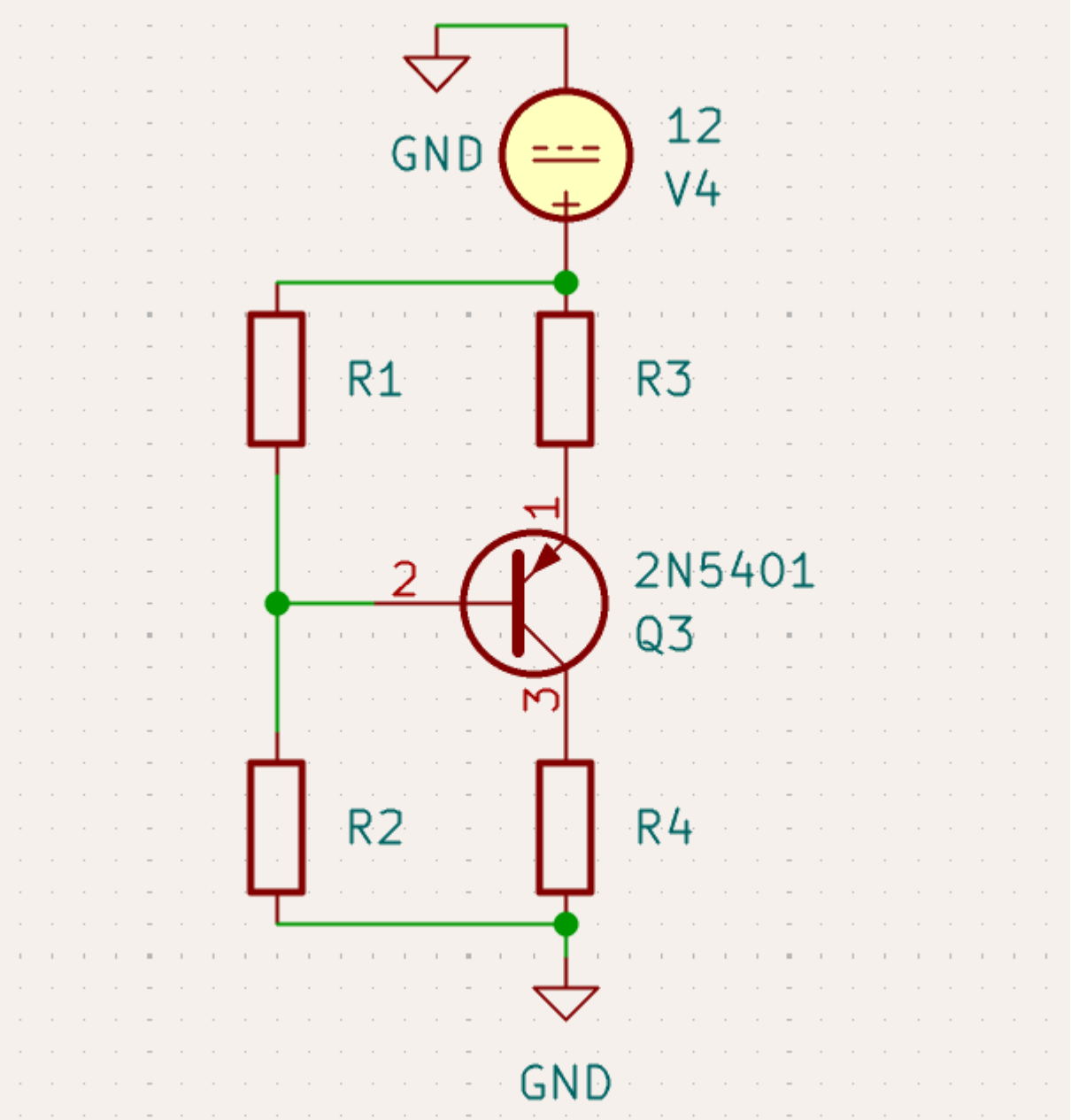



*Fig. 6. Initial bias circuit of the PNP stage*

Roles of the resistors:

- $R_1$ — $V_{CC}$ → base (top of the divider)
- $R_2$ — base → GND (bottom of the divider)
- $R_3$ — $V_{CC}$ → emitter (emitter resistor)
- $R_4$ — collector → GND (collector resistor)

**$R_3$ (emitter resistor).** The voltage across $R_3$ is taken as $10\ \%$ of $V_{CC}$. This is a compromise between operating-point stability and output swing.

$$V_{R3} = \frac{V_{CC}}{10} = \frac{12}{10} = 1.2\ \text{V}$$

$$I_E = I_B + I_C = 0.04\ \text{mA} + 8.5\ \text{mA} = 8.54\ \text{mA}, \qquad I_{R3} = I_E$$

$$R_3 = \frac{V_{R3}}{I_{R3}} = \frac{1.2}{0.00854} \approx 140.5\ \Omega\ \rightarrow\ 150\ \Omega\ (\text{E24})$$

**$R_4$ (collector resistor).**

$$V_{R4} = V_{CC} - V_{R3} - V_{CE} = 12 - 1.2 - 6 = 4.8 \text{ V}, \qquad I_{R4} = I_C = 8.5 \text{ mA}$$

$$R_4 = \frac{V_{R4}}{I_{R4}} = \frac{4.8}{0.0085} \approx 564.7 \ \Omega \ \rightarrow \ 560 \ \Omega \ (\text{E24})$$

**Base voltage.**

$$V_E = V_{CC} - V_{R3} = 12 - 1.2 = 10.8 \text{ V}, \qquad V_B = V_E - V_{EB} = 10.8 - 0.69 = 10.11 \text{ V}$$

**$R_1$.** The divider current is taken as $10\, I_B$.

$$V_{R1} = V_{CC} - V_B = 12 - 10.11 = 1.89 \text{ V}, \qquad I_{R1} = 10 \cdot I_B = 10 \cdot 0.00004 = 0.0004 \text{ A} = 400 \ \mu\text{A}$$

$$R_1 = \frac{V_{R1}}{I_{R1}} = \frac{1.89}{0.0004} = 4725 \ \Omega \ \rightarrow \ 4.7 \ \text{k}\Omega \ (\text{E24})$$

**$R_2$.**

$$V_{R2} = V_{CC} - V_{R1} = 12 - 1.89 = 10.11 \text{ V}, \qquad I_{R2} = I_{R1} + I_B = 0.0004 + 0.00004 = 0.00044 \text{ A}$$

$$R_2 = \frac{V_{R2}}{I_{R2}} = \frac{10.11}{0.00044} \approx 22.98 \ \text{k}\Omega \ \rightarrow \ 22 \ \text{k}\Omega \ (\text{E24})$$

**Summary:**

| Resistor | Calculated | E24 |
|---|---|---|
| $R_1$ | $4725 \ \Omega$ | **$4.7 \ \text{k}\Omega$** |
| $R_2$ | $22.98 \ \text{k}\Omega$ | **$22 \ \text{k}\Omega$** |
| $R_3$ | $140.5 \ \Omega$ | **$150 \ \Omega$** |
| $R_4$ | $564.7 \ \Omega$ | **$560 \ \Omega$** |

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# 6. DC Simulation

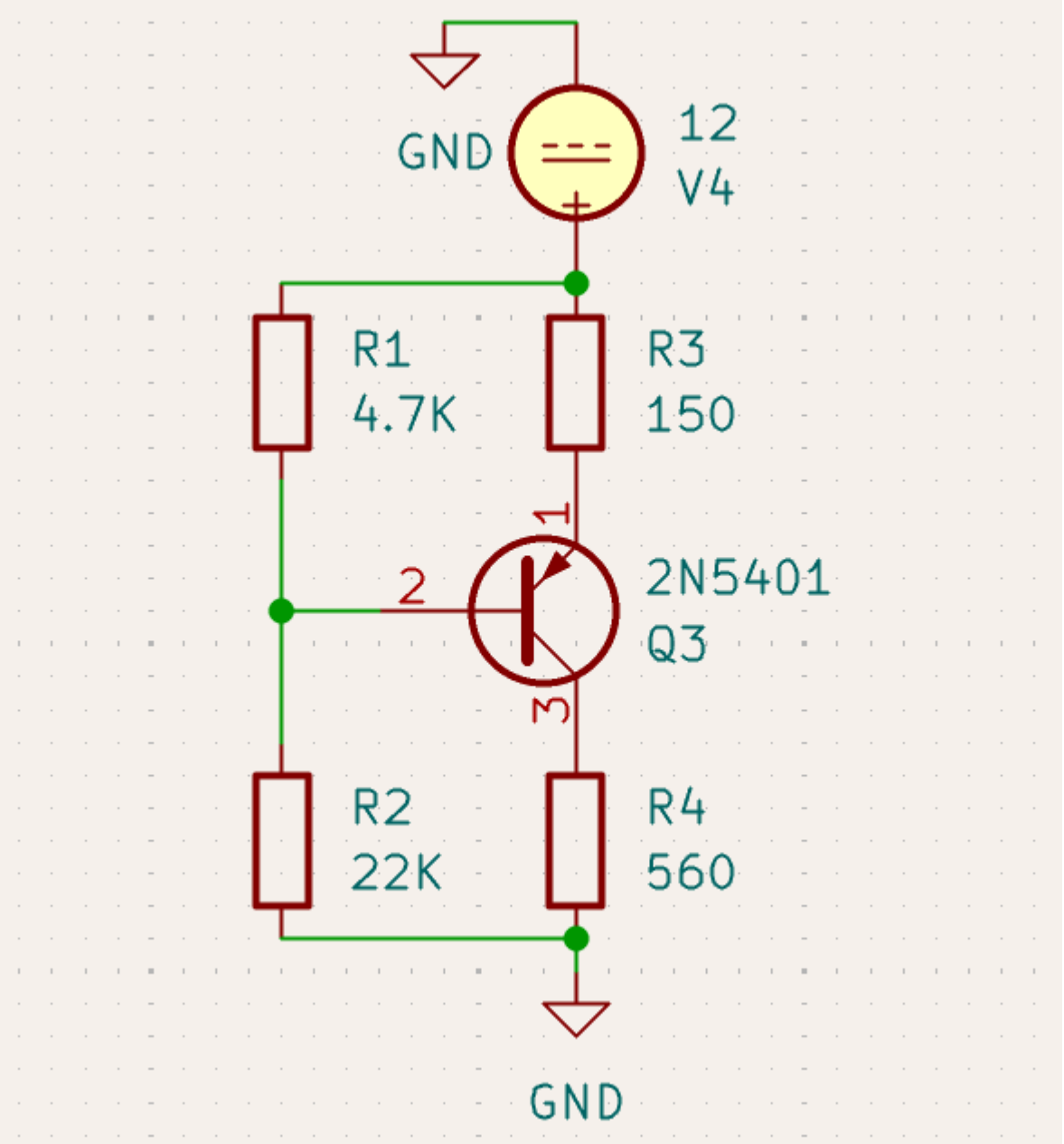



*Fig. 7. Bias circuit with the E24 resistor values ($V_{CC} = 12\ \mathrm{V}$)*

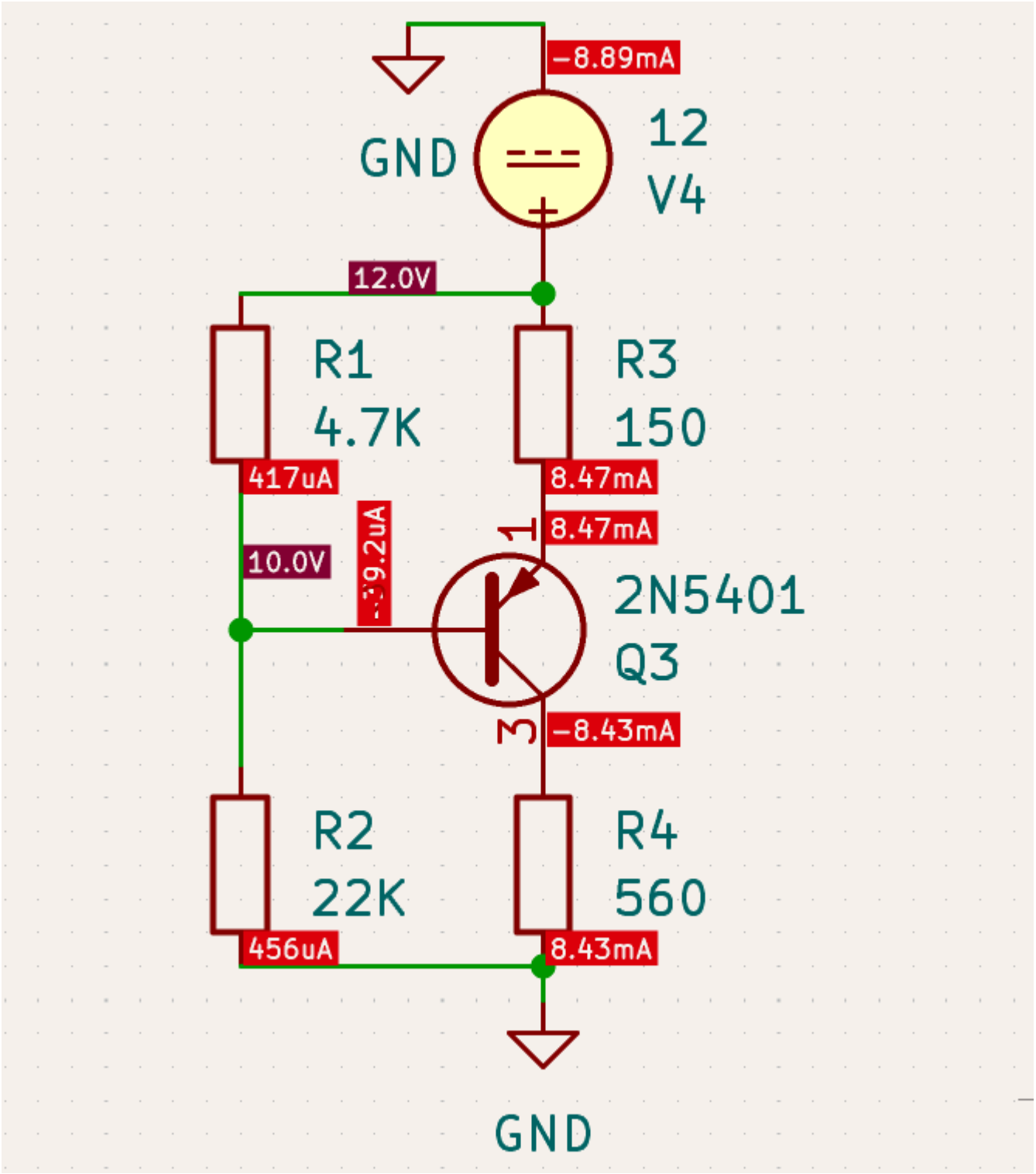



*Fig. 8. DC operating point simulation results*

| Quantity | Chosen | Simulated |
|---|---|---|
| $\lvert I_C \rvert$ | 8.5 mA | **8.431 mA** |
| $\lvert I_B \rvert$ | 40 $\mu$A | **39.19 $\mu$A** |
| $V_E$ | 10.8 V | **10.730 V** |
| $V_B$ | 10.11 V | **10.039 V** |
| $V_C$ | 4.8 V | **4.721 V** |

| Quantity | Chosen | Simulated |
|---|---|---|
| $V_{CE} = V_E - V_C$ | $6\ \text{V}$ | **$6.008\ \text{V}$** |
| $V_{EB} = V_E - V_B$ | $0.69\ \text{V}$ | **$0.690\ \text{V}$** |
| $\beta = I_C / I_B$ | $212$ | **$215$** |
| $P_Q$ | $51\ \text{mW}$ | **$50.7\ \text{mW}$** |

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## 7. Error Calculation

$$\delta = \frac{|X_{sim} - X_{chosen}|}{X_{chosen}} \cdot 100\%$$

$$\delta_{I_C} = \frac{|8.431 - 8.5|}{8.5} \cdot 100\% = 0.81\%, \qquad \delta_{V_{CE}} = \frac{|6.008 - 6|}{6} \cdot 100\% = 0.14\%$$

$$\delta_{I_B} = \frac{|39.19 - 40|}{40} \cdot 100\% = 2.0\%, \qquad \delta_{V_{EB}} = \frac{|0.690 - 0.69|}{0.69} \cdot 100\% \approx 0\%$$

**Possible causes of the errors:**

1. Rounding of the resistor values to the standard E24 series.
2. $\beta$ and $V_{EB}$ were read from the graphs. At the actual operating point the model gives $\beta \approx 215$ instead of $212$.
3. Rounding of intermediate values during the calculations.

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## 8. Conclusions

1. For variant $N = 6$ ($n_1 = 6, n_2 = 12$), the PNP transistor 2N5401 with supply voltage $V_{CC} = 12\ \text{V}$ was studied.
2. The input and output I–V characteristics were simulated in KiCad (ngspice) within the safe operating area. The model gives $\beta \approx 205 \ldots 215$ (datasheet: $60 \ldots 240$) and $V_{EB} \approx 0.69\ \text{V}$ at $I_C \approx 8.5\ \text{mA}$.
3. The operating point $I_C = 8.5\ \text{mA}$, $V_{CE} = 6\ \text{V}$ was chosen in the middle of the load line, which gives the maximum symmetric output swing. It dissipates $51\ \text{mW}$, far below $P_{tot} = 625\ \text{mW}$.
4. The bias resistors were calculated and rounded to the E24 series: $R_1 = 4.7\ \text{k}\Omega$, $R_2 = 22\ \text{k}\Omega$, $R_3 = 150\ \Omega$, $R_4 = 560\ \Omega$.
5. The DC simulation matches the chosen operating point: the error is $0.81\ \%$ for $I_C$ and $0.14\ \%$ for $V_{CE}$. The main sources of error are E24 rounding and approximate parameters read from the graphs.